

NetSage AI

Applied AI + Network Troubleshooting

Cisco-AICTE Virtual Internship Program 2026 — Project 2

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Agenda

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About the Internship Program

The Cisco-AICTE Virtual Internship Program partners with AICTE to give engineering students hands-on exposure to networking and applied-AI challenges using Cisco's curriculum and Packet Tracer.

Students take on industry-style problem statements — individually or in teams — evaluated by a Cisco review team against strict submission and originality guidelines.

2026

Program Year

Project 2

Applied AI + Network Troubleshooting

Problem Statement

“A PC gets an IP address but can't reach a server — is it VLAN, routing, DHCP, DNS, ACL, or NAT?”

Junior engineers know individual commands but struggle to connect a symptom to its true root cause across OSI layers. NetSage AI builds a troubleshooting assistant that reasons from symptoms and show-command evidence to a likely fault, OSI layer, next command, and fix — always reviewed by a human before being accepted.

Objectives

01

Build a realistic 30-case troubleshooting dataset across 8 fault categories

02

Design a structured AI prompt forcing evidence-grounded JSON diagnoses

03

Implement a deterministic rule checker as a non-AI safety net

04

Enforce mandatory human review of every AI diagnosis

05

Summarize outcomes in a measurable dashboard

Tools & Technologies

Packet Tracer

Lab topology + fault simulation

Python

Deterministic rule checker

AI Prompting

Structured JSON diagnosis

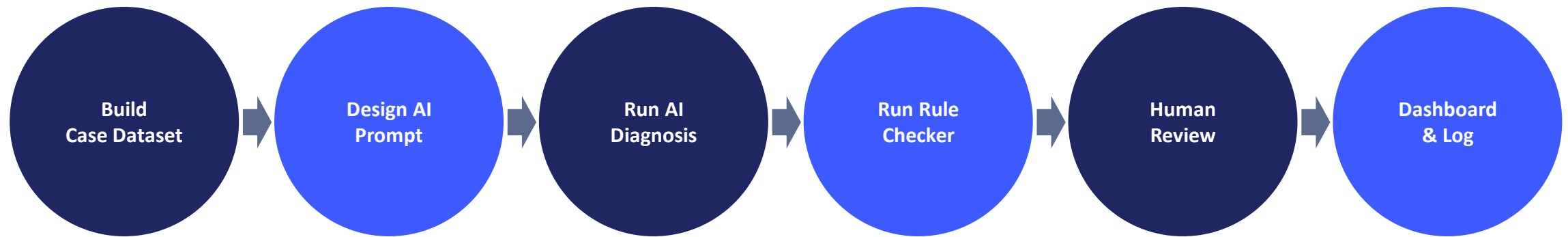
Excel

Dashboard & charts

CSV / Markdown

Case data & review logs

Methodology



Every diagnosis passes through human review before being treated as a fix.

Case Dataset

30

Troubleshooting Cases

8

Fault Categories

37%

Caught by Rule Checker

VLAN · Gateway · DHCP · DNS · Routing · ACL · NAT · Wireless

Each case includes: symptom, topology note, show-command output, expected fault, OSI layer, concept tag, and severity.

AI Prompt Design

The prompt forces the AI to reason only from given evidence and return a fixed JSON schema:

root_cause

osi_layer

confidence

evidence

next_command

fix_steps

```
{  
  
  "root_cause": "Fa0/3 in VLAN 20,  
  not VLAN 10",  
  
  "osi_layer": "Layer 2",  
  
  "confidence": "high",  
  
  "evidence": "show vlan brief  
  lists Fa0/3 under VLAN 20",  
  
  "next_command": "show interfaces  
  Fa0/3 switchport",  
  
  "fix_steps": [ "switchport  
  access vlan 10", "... " ]  
  
}
```

Two Layers of Safety

Deterministic Rule Checker

- Duplicate IPs
- Wrong subnet / wildcard masks
- Gateway mismatches
- Interfaces administratively down
- Inactive / missing VLANs
- Missing routes, ACL & NAT errors

Runs independently of the AI — catches 11 of 30 cases (37%) from text patterns alone.

Mandatory Human Review

Every AI diagnosis is marked:

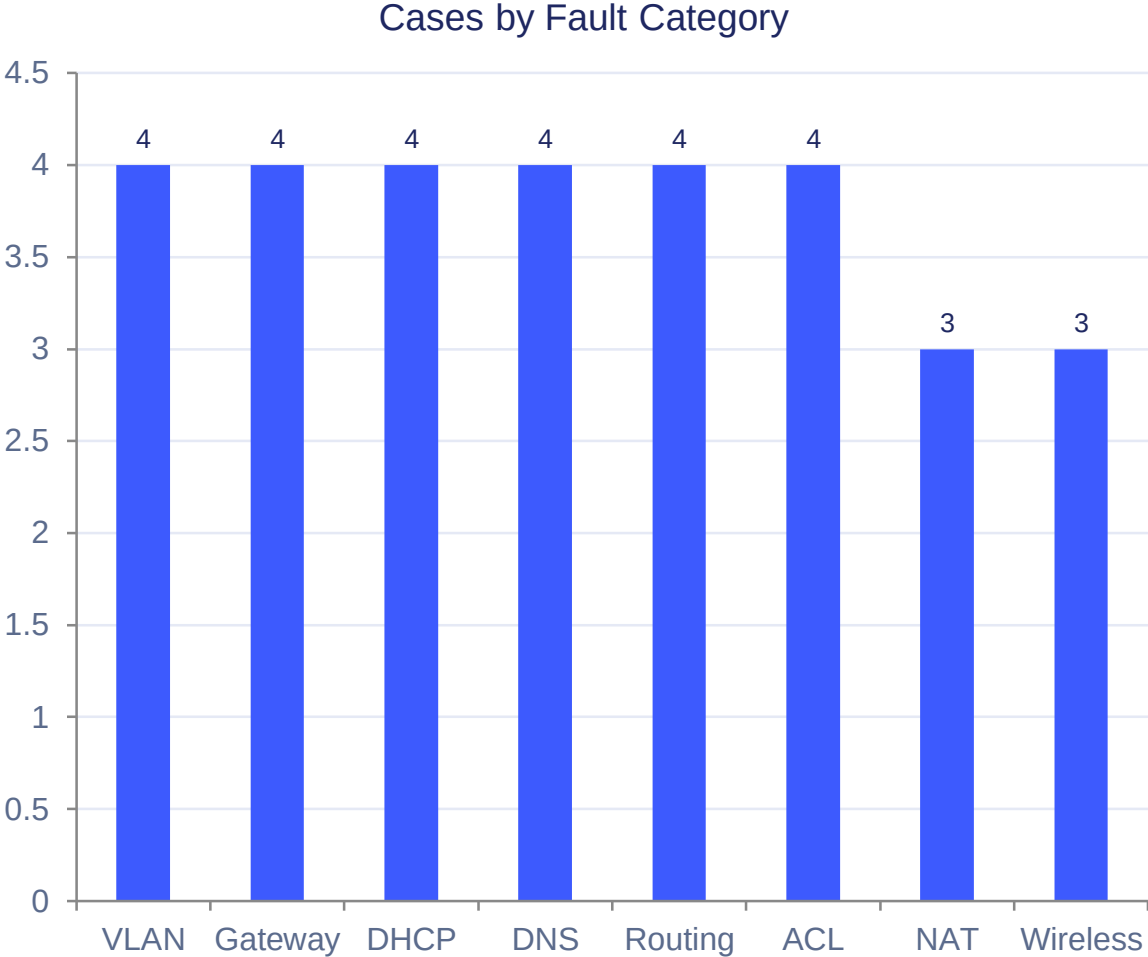
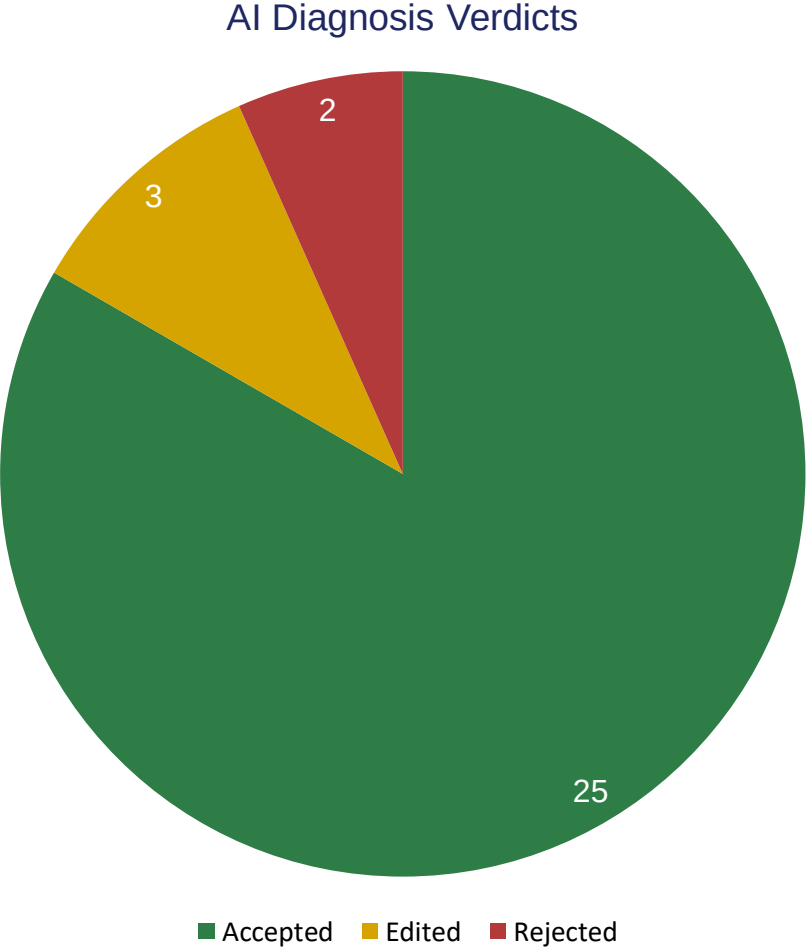
✓ **Accepted — 25 cases (83%)**

✎ **Edited — 3 cases**

✗ **Rejected — 2 cases**

No diagnosis is ever applied to the network without sign-off.

Results Dashboard



Responsible AI: Learning from Mistakes

C04 — Rejected

AI blamed the wrong VLAN field; real cause was a missing voice VLAN config.

C18 — Rejected

AI guessed a timer mismatch; evidence actually showed a classic MTU-mismatch (EXSTART) signature.

C07 — Edited

AI assumed OSPF was involved in a topology that had no routing protocol running at all.

5 of 30 cases required human correction (83% first-pass agreement) — each was fed back into refining the AI prompt.

Key Learnings

- Strengthened Layer 2/3 troubleshooting across VLAN, routing, DHCP, DNS, ACL, NAT, and wireless
- Learned to design AI prompts that force evidence-grounded, structured output
- Understood why AI alone is not enough — deterministic checks + human review reduce risk
- Built a dashboard that makes AI performance measurable and auditable
- Practiced documenting AI failures transparently, not just successes

Conclusion

NetSage AI combined classical network troubleshooting with an applied AI workflow under a strict human-in-the-loop safety rule — from dataset and prompt design through rule-checking, human review, and reporting. Completed individually, start to finish.

Thank You

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